

Mathematics A

Unit 2 • Chapter OL-T15 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 19 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T15 • Expertise Q16+ • 5 items • 19 marks

Chapter	Items	Marks	Starts
01. OL-T15 Direct & Inverse Proportion	5	19	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Direct & Inverse Proportion

Unit 2 • Expertise • 19 marks • 5 item(s)

June 2025 | 4MA1-1H Q18 | 4 marks

Recover the input under inverse variation

June 2023 | 4MA1-2H Q16 | 5 marks

Root direct variation $y = k\sqrt{x}$

June 2022 | 4MA1-2H Q17 | 4 marks

Power direct variation $y = kx^n$

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

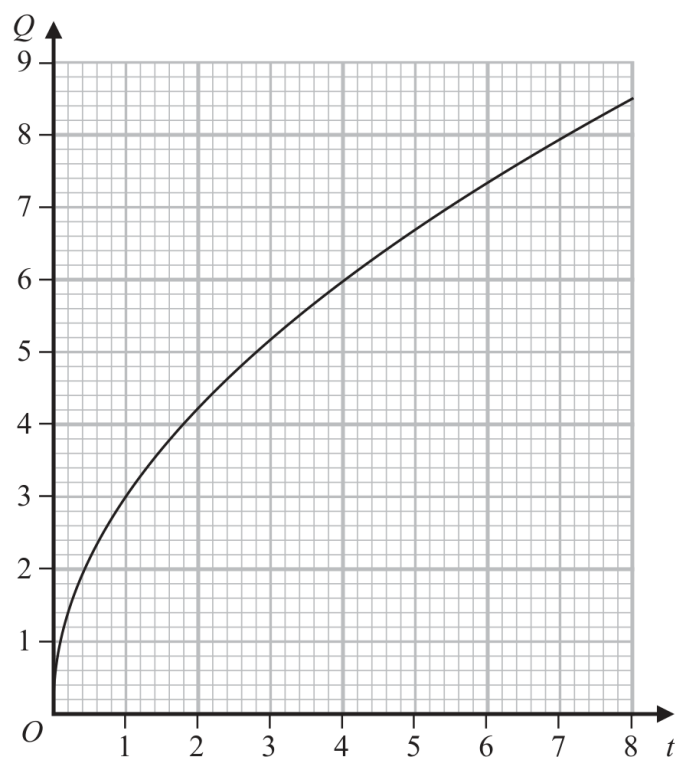
WORKING SPACE

4MA1-2H Q16

5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

..... %
(2)

Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q17 Q17

4 marks

- 17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

WORKING SPACE

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

YOUR WORKING

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

YOUR WORKING
